

```

import React, {
  forwardRef,
  useState,
  useEffect,
  useRef,
  useCallback,
  useMemo,
} from "react";
import { clsx, type ClassValue } from "clsx";
import { twMerge } from "tailwind-merge";
import { ArrowUp, ArrowDown, ArrowUpDown } from "lucide-react";

// 🕸 THE SACRED OFFICIAL TANSTACK CORE & VIRTUAL IMPORTERS SECURED!
import {
  createTable,
  type ColumnDef,
  type RowSelectionState,
  type SortingState,
  type ColumnSizingState,
  type Row,
  type Column,
  getCoreRowModel,
  getSortedRowModel,
  getFilteredRowModel,
} from "@tanstack/table-core";
import { useVirtualizer } from "@tanstack/react-virtual";

import { TextBox } from "../data-entry/text-box";
import { NumericEdit } from "../data-entry/numeric-edit";
import { DateEdit } from "../data-entry/date-edit";
import { PremiumCheckbox } from "../selector/checkbox";
import { ComboBox } from "../selector/combo-box";
import { HintBox } from "../feedback/hint-box";

export function cn(...inputs: ClassValue[]) {
  return twMerge(clsx(inputs));
}

export interface PopupMenuConfig {
  trigger: (e: React.MouseEvent, recordRow?: Record<string, unknown>) => void;
  [key: string]: unknown;
}

export interface GridColumnLayout {
  key: string;
  headerLabel: string;
  widthPercent: number;
  align?: "left" | "center" | "right";
  editType?: "text" | "numeric" | "date" | "checkbox" | "combo";
  editOptions?: Record<string, unknown>[];
}

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    enableSorting?: boolean;
    enableFiltering?: boolean;
}

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export interface GridDataChangePayload {
  rowIndex: number;
  columnKey: string;
  oldValue: unknown;
  newValue: unknown;
}

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export interface GridProps extends Omit<
  React.HTMLAttributes<HTMLDivElement>,
  "onChange"
> {
  label?: string;
  error?: string;
  hint?: string;
  popupMenu?: PopupMenuConfig;
  columns: GridColumnLayout[];
  data: Record<string, unknown>[];
  showSelectionCheckbox?: boolean;
  isEditable?: boolean;
  isRowMovable?: boolean;
  isColumnMovable?: boolean;
  rowIdentifierKey?: string;
  rowHeightPx?: number;
  activeRowKey?: string;
  onRowClick?: (row: Record<string, unknown>, index: number) => void;
  onDataChange?: (
    updatedData: Record<string, unknown>[],
    changedInfo: GridDataChangePayload,
  ) => void;
  onRowMove?: (fromIndex: number, toIndex: number) => void;
  onColumnMove?: (fromIndex: number, toIndex: number) => void;
}

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export const Grid = forwardRef<HTMLDivElement, GridProps>(
  (
    {
      label,
      error,
      hint,
      popupMenu,
      columns = [],
      data = [],
      showSelectionCheckbox = false,
      isEditable = false,
      isRowMovable = false,
      isColumnMovable = false,

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    rowIdentifierKey = "id",
    rowHeightPx = 42,
    activeRowKey,
    onRowClick,
    onDataChange,
    onRowMove,
    onColumnMove,
    className,
    ...props
  },
  ref,
) => {
  const parentRef = useRef<HTMLDivElement>(null);
  const tableRef = useRef<HTMLTableElement>(null);
  const [sorting, setSorting] = useState<SortingState>([]);
  const [columnFilters, setColumnFilters] = useState<
    import("@tanstack/table-core").ColumnFiltersState
  >([]);
  const [globalFilter, setGlobalFilter] = useState<string>("");
  const [columnOrder, setColumnOrder] = useState<string[]>([]);
  const [columnVisibility, setColumnVisibility] = useState<
    Record<string, boolean>
  >({});
  const [rowSelection, setRowSelection] = useState<RowSelectionState>({});
  const [editingCell, setEditingCell] = useState<{
    rowIndex: number;
    columnKey: string;
  } | null>(null);
  const [editValue, setEditValue] = useState<unknown>("");
  const [columnSizing, setColumnSizing] = useState<ColumnSizingState>({});
  const [isResizing, setIsResizing] = useState<{
    columnKey: string;
    startX: number;
    startWidth: number;
  } | null>(null);
  const [draggedColumn, setDraggedColumn] = useState<string | null>(null);
  const [draggedRow, setDraggedRow] = useState<number | null>(null);
  const [filterInputs, setFilterInputs] = useState<Record<string, string>>(
    {},
  );
};

const columnDefs = useMemo(() : ColumnDef<Record<string, unknown>>[] => {
  const defs: ColumnDef<Record<string, unknown>>[] = [];

  if (showSelectionCheckbox) {
    defs.push({
      id: "__selection__",
      header: ({ table }) => (
        <PremiumCheckbox
          checked={table.getIsAllRowsSelected()}

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        isIndeterminate={table.getIsSomeRowsSelected()}
        onChange={(checked) => table.toggleAllRowsSelected(!checked)}
        className="h-4 w-4"
      />
    ),
    cell: ({ row }) => (
      <PremiumCheckbox
        checked={row.getIsSelected()}
        onChange={(checked) => row.toggleSelected(!checked)}
        className="h-4 w-4"
      />
    ),
    size: 48,
    enableSorting: false,
    enableColumnFilter: false,
    enableResizing: false,
  });
}

columns.forEach((col) => {
  defs.push({
    id: col.key,
    accessorKey: col.key,
    header: col.headerLabel,
    size: Math.max(80, (col.widthPercent / 100) * 1200),
    minSize: 60,
    maxSize: 800,
    enableSorting: col.enableSorting !== false,
    enableColumnFilter: col.enableFiltering !== false,
    enableResizing: true,
    enableHiding: true,
  });
});

return defs;
}, [columns, showSelectionCheckbox]);

const table = useMemo(() => {
  const tableInstance = createTable<Record<string, unknown>>>({
    data,
    columns: columnDefs,
    state: {
      sorting,
      columnFilters,
      globalFilter,
      columnOrder,
      columnVisibility,
      rowSelection,
      columnSizing,
    },
  });

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    onSortingChange: setSorting,
    onColumnFiltersChange: setColumnFilters,
    onGlobalFilterChange: setGlobalFilter,
    onColumnOrderChange: setColumnOrder,
    onColumnVisibilityChange: setColumnVisibility,
    onRowSelectionChange: setRowSelection,
    onColumnSizingChange: setColumnSizing,
    getRowId: (row) =>
      String(row[rowIdentifierKey] ?? row.id ?? Math.random()),
    manualSorting: true,
    manualFiltering: true,
    enableRowSelection: true,
    getCoreRowModel: getCoreRowModel(),
    getSortedRowModel: getSortedRowModel(),
    getFilteredRowModel: getFilteredRowModel(),
    onStateChange: () => {},
    renderFallbackValue: () => null,
  });
  return tableInstance;
}, [
  data,
  columnDefs,
  sorting,
  columnFilters,
  globalFilter,
  columnOrder,
  columnVisibility,
  rowSelection,
  columnSizing,
  rowIdentifierKey,
]);

const virtualizer = useVirtualizer({
  count: table.getRowModel().rows.length,
  getScrollElement: () => parentRef.current,
  estimateSize: () => rowHeightPx,
  overscan: 5,
});

const handleEditChange = useCallback((newValue: unknown) => {
  setEditValue(newValue);
}, []);

const handleEditBlur = useCallback(
  (rowIndex: number, columnKey: string, oldValue: unknown) => {
    if (
      editingCell &&
      editingCell.rowIndex === rowIndex &&
      editingCell.columnKey === columnKey
    ) {

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    if (editValue !== oldValue) {
      const updatedData = [...data];
      updatedData[rowIndex] = {
        ...updatedData[rowIndex],
        [columnKey]: editValue,
      };

      onDataChange?.(updatedData, {
        rowIndex,
        columnKey,
        oldValue,
        newValue: editValue,
      });
    }
    setEditingCell(null);
    setEditValue("");
  }
},
[editingCell, editValue, data, onDataChange],
);

const handleKeyDown = useCallback(
  (
    e: React.KeyboardEvent,
    rowIndex: number,
    columnKey: string,
    value: unknown,
  ) => {
    if (e.key === "Enter" && !e.shiftKey) {
      e.preventDefault();
      if (!isEditable) return;
      setEditingCell({ rowIndex, columnKey });
      setEditValue(value);
    } else if (e.key === "Escape" && editingCell) {
      setEditingCell(null);
      setEditValue("");
    } else if (e.key === "Tab") {
      if (editingCell) {
        handleEditBlur(
          editingCell.rowIndex,
          editingCell.columnKey,
          data[editingCell.rowIndex]?.[editingCell.columnKey] as unknown,
        );
      }
    }
  },
  [isEditable, editingCell, data, handleEditBlur],
);

const handleRowClick = useCallback(

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    (row: Record<string, unknown>, index: number) => {
      onRowClick?.(row, index);
    },
    [onRowClick],
  );

  const handleContextMenu = useCallback(
    (e: React.MouseEvent, row: Record<string, unknown>) => {
      e.preventDefault();
      popupMenu?.trigger(e, row);
    },
    [popupMenu],
  );

  const handleSort = useCallback(
    (column: Column<Record<string, unknown>>) => {
      const isSorted = column.getIsSorted();
      column.toggleSorting(
        isSorted === "asc" ? false : isSorted === "desc" ? undefined : true,
      );
    },
    [],
  );

  const handleFilterChange = useCallback(
    (columnKey: string, value: string) => {
      setFilterInputs((prev) => ({ ...prev, [columnKey]: value }));
      table.setColumnFilters((prev) => {
        const existing = prev.find((f) => f.id === columnKey);
        if (value === "") {
          return prev.filter((f) => f.id !== columnKey);
        }
        if (existing) {
          return prev.map((f) => (f.id === columnKey ? { ...f, value } : f));
        }
        return [...prev, { id: columnKey, value }];
      });
    },
    [table],
  );

  const handleColumnResizeStart = useCallback(
    (e: React.MouseEvent, columnKey: string) => {
      e.preventDefault();
      e.stopPropagation();
      const column = table.getColumn(columnKey);
      if (column) {
        setIsResizing({
          columnKey,
          startX: e.clientX,

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        startWidth: column.getSize(),
    });
}
},
[table],
);

const handleColumnResize = useCallback(
    (e: MouseEvent) => {
        if (isResizing) {
            const newWidth = Math.max(
                60,
                isResizing.startWidth + (e.clientX - isResizing.startX),
            );
            table.setColumnSizing((prev) => ({
                ...prev,
                [isResizing.columnKey]: newWidth,
            }));
        }
    },
    [isResizing, table],
);

const handleColumnResizeEnd = useCallback(() => {
    setIsResizing(null);
}, []);

const handleDragStart = useCallback(
    (e: React.DragEvent, columnKey: string) => {
        if (!isColumnMovable) return;
        setDraggedColumn(columnKey);
        e.dataTransfer.effectAllowed = "move";
    },
    [isColumnMovable],
);

const handleDragOver = useCallback((e: React.DragEvent) => {
    e.preventDefault();
    e.dataTransfer.dropEffect = "move";
}, []);

const handleDrop = useCallback(
    (e: React.DragEvent, targetColumnKey: string) => {
        e.preventDefault();
        if (draggedColumn && draggedColumn !== targetColumnKey) {
            const columns = table.getAllColumns();
            const fromIndex = columns.findIndex(
                (c: Column<Record<string, unknown>>) => c.id === draggedColumn,
            );
            const toIndex = columns.findIndex(

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        (c: Column<Record<string, unknown>>) => c.id === targetColumnKey,
    );
    if (fromIndex !== -1 && toIndex !== -1) {
        const newOrder = [...columnOrder];
        const [removed] = newOrder.splice(fromIndex, 1);
        newOrder.splice(toIndex, 0, removed);
        setColumnOrder(newOrder);
        onColumnMove?.(fromIndex, toIndex);
    }
    }
    setDraggedColumn(null);
},
[draggedColumn, columnOrder, table, onColumnMove],
);

const handleRowDragStart = useCallback(
    (e: React.DragEvent, rowIndex: number) => {
        if (!isRowMovable) return;
        setDraggedRow(rowIndex);
        e.dataTransfer.effectAllowed = "move";
    },
    [isRowMovable],
);

const handleRowDragOver = useCallback((e: React.DragEvent) => {
    e.preventDefault();
    e.dataTransfer.dropEffect = "move";
}, []);

const handleRowDrop = useCallback(
    (e: React.DragEvent, targetRowIndex: number) => {
        e.preventDefault();
        if (draggedRow !== null && draggedRow !== targetRowIndex) {
            onRowMove?.(draggedRow, targetRowIndex);
        }
        setDraggedRow(null);
    },
    [draggedRow, onRowMove],
);

useEffect(() => {
    const handleMouseMove = (e: MouseEvent) => handleColumnResize(e);
    const handleMouseUp = () => handleColumnResizeEnd();

    if (isResizing) {
        window.addEventListener("mousemove", handleMouseMove);
        window.addEventListener("mouseup", handleMouseUp);
    }

    return () => {

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        window.removeEventListener("mousemove", handleMouseMove);
        window.removeEventListener("mouseup", handleMouseUp);
    };
}, [isResizing, handleColumnResize, handleColumnResizeEnd]);

const sortedColumns = useMemo(() => {
    const allColumns = table.getAllColumns();
    if (columnOrder.length > 0) {
        return columnOrder
            .map((id) =>
                allColumns.find(
                    (c: Column<Record<string, unknown>>) => c.id === id,
                ),
            )
            .filter(Boolean) as Column<Record<string, unknown>>[];
    }
    return allColumns;
}, [table, columnOrder]);

const headerGroups = table.getHeaderGroups();
const rows = table.getRowModel().rows;

const getAlignmentClass = (align?: string) => {
    switch (align) {
        case "center":
            return "text-center";
        case "right":
            return "text-right";
        default:
            return "text-left";
    }
};

const renderCellEditor = (
    column: Column<Record<string, unknown>>,
    row: Row<Record<string, unknown>>,
    value: unknown,
) => {
    const colConfig = columns.find((c) => c.key === column.id);
    if (!colConfig || !colConfig.editType) return null;

    const isEditing =
        editingCell?.rowIndex === row.index &&
        editingCell?.columnKey === column.id;

    if (!isEditing) return null;

    const commonProps = {
        value: editValue,
        onChange: handleEditChange,
    };

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    onBlur: () => handleEditBlur(row.index, column.id, value),
    onKeyDown: (e: React.KeyboardEvent) =>
        handleKeyDown(e, row.index, column.id, value),
    className: "w-full h-full min-h-[36px]",
    error: error,
};

switch (colConfig.editType) {
    case "text":
        return <TextBox {...commonProps} value={String(editValue ?? "")} />;
    case "numeric":
        return (
            <NumericEdit {...commonProps} value={Number(editValue ?? 0)} />
        );
    case "date":
        return <DateEdit {...commonProps} value={String(editValue ?? "")} />;
    case "checkbox":
        return (
            <PremiumCheckbox
                checked={editValue as boolean}
                onChange={(checked) => handleEditChange(checked)}
                className="h-4 w-4"
            />
        );
    case "combo":
        return (
            <ComboBox
                options={colConfig.editOptions || []}
                value={String(editValue ?? "")}
                onChange={handleEditChange}
                className="w-full h-full min-h-[36px]"
            />
        );
    default:
        return null;
}
};

const renderCellContent = (
    column: Column<Record<string, unknown>>,
    row: Row<Record<string, unknown>>,
) => {
    const value = row.getValue(column.id);
    const colConfig = columns.find((c) => c.key === column.id);
    const isEditing =
        editingCell?.rowIndex === row.index &&
        editingCell?.columnKey === column.id;

    if (isEditing) {
        return renderCellEditor(column, row, value);
    }

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    }

    if (colConfig?.editType === "checkbox") {
      return (
        <PremiumCheckbox
          checked={value as boolean}
          disabled={!isEditable}
          onChange={
            isEditable
              ? (checked) => {
                  const updatedData = [...data];
                  updatedData[row.index] = {
                    ...updatedData[row.index],
                    [column.id]: checked,
                  };
                  onDataChange?.(updatedData, {
                    rowIndex: row.index,
                    columnKey: column.id,
                    oldValue: value,
                    newValue: checked,
                  });
                }
              : undefined
            }
          className="h-4 w-4"
        />
      );
    }

    return (
      <div className={cn("truncate", getAlignmentClass(colConfig?.align))}>
        {String(value ?? "")}
      </div>
    );
  };

  return (
    <div
      ref={(el) => {
        parentRef.current = el;
        if (ref) {
          if (typeof ref === "function") ref(el);
          else ref.current = el;
        }
      }}
      className={cn(
        "relative w-full overflow-hidden rounded-lg border
border-[color-mix(in_oklch,var(--color-border)_40%,transparent)] bg-card",
        className,
      )}
    >

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    {...props}
  >
    {hint && (
      <HintBox content={hint} className="mb-1.5">
        <span className="text-sm text-text-muted" />
      </HintBox>
    )}

    {label && (
      <div className="px-4 py-2 border-b
border-[color-mix(in_oklch,var(--color-border)_40%,transparent)] bg-card/40">
        <span className="font-semibold text-text-main">{label}</span>
      </div>
    )}

    <div
      className="relative overflow-auto"
      style={{ maxHeight: "600px" }}
      onKeyDown={(e) => {
        if (e.key === "ArrowDown" || e.key === "ArrowUp") {
          e.preventDefault();
        }
      }}
    >
      <table
        ref={tableRef}
        className="w-full border-collapse"
        style={{
          minWidth: sortedColumns.reduce(
            (sum: number, col: Column<Record<string, unknown>>) =>
              sum + (col.getSize() || 100),
            0,
          ),
        }}
        role="grid"
      >
        <thead className="sticky top-0 z-10">
          {headerGroups.map(
            (
              headerGroup: import("@tanstack/table-core").HeaderGroup<
                Record<string, unknown>
              >,
            ) => (
              <tr
                key={headerGroup.id}
                className="bg-card/40 border-b
border-[color-mix(in_oklch,var(--color-border)_40%,transparent)]"
              >
                {headerGroup.headers.map(
                  (

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        header: import("@tanstack/table-core").Header<
          Record<string, unknown>,
          unknown
        >,
      ) => {
        const column = header.column;
        const isSorted = column.getIsSorted();
        const canSort = column.getCanSort();
        const isResizing = column.getIsResizing();
        const isDragged = draggedColumn === column.id;

        return (
          <th
            key={column.id}
            className={cn(
              "relative px-3 py-2 font-semibold text-text-main
whitespace-nowrap",
              "border-r
border-[color-mix(in_oklch,var(--color-border)_40%,transparent)]",
              "last:border-r-0",
              "select-none",
              isDragged && "opacity-50",
              isResizing && "bg-brand-primary/10",
            )}
            style={{ width: column.getSize() }}
            draggable={
              isColumnMovable && column.id !== "__selection__"
            }
            onDragStart={(e) => handleDragStart(e, column.id)}
            onDragOver={handleDragOver}
            onDrop={(e) => handleDrop(e, column.id)}
            onMouseDown={(e) => {
              if (
                e.target === e.currentTarget &&
                column.getCanResize()
              ) {
                handleColumnResizeStart(e, column.id);
              }
            }}
          >
            <div className="flex items-center gap-2">
              {column.id === "__selection__" ? (
                typeof header.column.columnDef.header ===
                  "function" ? (
                    header.column.columnDef.header({
                      table,
                      column: header.column,
                      header: header.column.columnDef.header,
                    } as unknown as
import("@tanstack/table-core").HeaderContext<

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        Record<string, unknown>,
        unknown
    >)
) : (
    header.column.columnDef.header
)
) : (
    <>
    {canSort && (
        <button
            onClick={() => handleSort(column)}
            className={cn(
                "flex items-center gap-1 p-1 rounded
hover:bg-brand-primary/10 transition-colors",
                isSorted && "text-brand-primary",
            )}
            aria-label={`Sort by ${column.id}`}
        >
            {typeof header.column.columnDef.header ===
"function"
                ? header.column.columnDef.header({
                    table,
                    column: header.column,
                    header:
                        header.column.columnDef.header,
                }) as unknown as
import("@tanstack/table-core").HeaderContext<
                Record<string, unknown>,
                unknown
            >)}
            : header.column.columnDef.header}
        {isSorted === "asc" && (
            <ArrowUp className="h-3.5 w-3.5" />
        )}
        {isSorted === "desc" && (
            <ArrowDown className="h-3.5 w-3.5" />
        )}
        {!isSorted && (
            <ArrowUpDown className="h-3.5 w-3.5
text-text-muted" />
        )}
    </button>
    )}
    {!canSort &&
        (typeof header.column.columnDef.header ===
"function"
            ? header.column.columnDef.header({
                table,
                column: header.column,
                header:

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        header.column.columnDef.header,
        } as unknown as
import("@tanstack/table-core").HeaderContext<
        Record<string, unknown>,
        unknown
        >)
        : header.column.columnDef.header)}}
    </>
  )}
  {column.getCanFilter() &&
    column.id !== "__selection__" && (
      <div className="relative flex-1 min-w-0">
        <input
          type="text"
          placeholder="Filter..."
          value={filterInputs[column.id] || ""}
          onChange={(e) =>
            handleFilterChange(
              column.id,
              e.target.value,
            )
          }
          onClick={(e) => e.stopPropagation()}
          className="w-full px-2 py-1 text-xs
bg-background border
border-[color-mix(in_oklch,var(--color-border)_40%,transparent)] rounded
focus:outline-none focus:ring-1 focus:ring-brand-primary"
        />
      </div>
    )}
  </div>
  {column.getCanResize() && (
    <div
      className="absolute right-0 top-0 h-full w-1
cursor-col-resize hover:w-2 hover:bg-brand-primary/30 transition-all"
      onMouseDown={(e) =>
        handleColumnResizeStart(e, column.id)
      }
    />
  )}
</th>
);
},
)}
</tr>
),
)}
</thead>
<tbody>
  {virtualizer.getVirtualItems().map((virtualRow) => {

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const row = rows[virtualRow.index];
const isSelected = row.getIsSelected();
const isActive =
  activeRowKey &&
  row.getValue(rowIdentifierKey) === activeRowKey;
const isEven = virtualRow.index % 2 === 0;

return (
  <tr
    key={row.id}
    className={cn(
      "transition-colors duration-150",
      "hover:bg-brand-primary/3",
      isSelected &&
        "bg-brand-primary/5 text-text-main font-semibold border-l-2
border-brand-primary",
      isActive && "bg-brand-primary/10",
      isEven &&

      "bg-[color-mix(in_oklch,var(--color-border)_5%,transparent)]",
      "border-b
border-[color-mix(in_oklch,var(--color-border)_20%,transparent)]",
      "last:border-b-0",
    )}
    style={{
      height: rowHeightPx,
      transform: `translateY(${virtualRow.start}px)`,
    }}
    draggable={isRowMovable}
    onDragStart={(e) => handleRowDragStart(e, virtualRow.index)}
    onDragOver={handleRowDragOver}
    onDrop={(e) => handleRowDrop(e, virtualRow.index)}
    onClick={() =>
      handleRowClick(row.original, virtualRow.index)
    }
    onContextMenu={(e) => handleContextMenu(e, row.original)}
  >
    {sortedColumns.map((column) => (
      <td
        key={column.id}
        className={cn(
          "px-3 py-2 whitespace-nowrap overflow-hidden",
          "border-r
border-[color-mix(in_oklch,var(--color-border)_20%,transparent)]",
          "last:border-r-0",
          getAlignmentClass(
            columns.find((c) => c.key === column.id)?.align,
          ),
        )}
        style={{ width: column.getSize() }}

```

```

        >
        {renderCellContent(column, row)}
      </td>
    )})
  </tr>
);
}}
{rows.length === 0 && (
  <tr>
    <td
      colSpan={sortedColumns.length}
      className="px-4 py-8 text-center text-text-muted"
    >
      No data available
    </td>
  </tr>
)}
</tbody>
</table>
</div>

{error && (
  <div className="px-4 py-2 border-t
border-[color-mix(in_oklch,var(--color-border)_40%,transparent)] bg-destructive/5">
    <span className="text-sm text-destructive">{error}</span>
  </div>
)}
</div>
);
},
);
Grid.displayName = "Grid";

```